B.Sc. Engg. CSE 4th Semester 05 March 2024 (Afternoon)

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION SUMMER SEMESTER, 2022-2023

DURATION: 1 HOUR 30 MINUTES FULL MARKS: 100

CSE 4405: Data and Telecommunications

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) Describe the three criteria necessary for effective and efficient networks. Mr. Umair and Mr. Uzair are talking over Cell Phones. Cellular communication uses the infrastructure provided by mobile operators which includes radio resources. Identify the five components of Data Communications in the given system. 4 + 4 (CO1) (PO1)

   b) Briefly explain the necessity of layering in designing a communication system. Match the followings to one or more layers of the OSI model and briefly explain the major responsibilities of the mentioned layers: 4 + 12 (CO1) (PO1)

      i. Access control
      ii. Encryption
      iii. Transmission mode
      iv. Segmentation and reassembly

   c) Write down the significance of different levels of addresses used in the TCP/IP protocol suite. In Figure 1, the logical address is represented by an English letter and the physical address is represented by a numerical value. Example: The top-left PC (sender) has logical address A and physical address 10. Consider that, the sender with a logical address A sends data to the receiver with logical address P. Find out the values of C, D, L, and Z. How would the values change when data is traveling in the opposite direction?. 10 (CO1) (PO1)

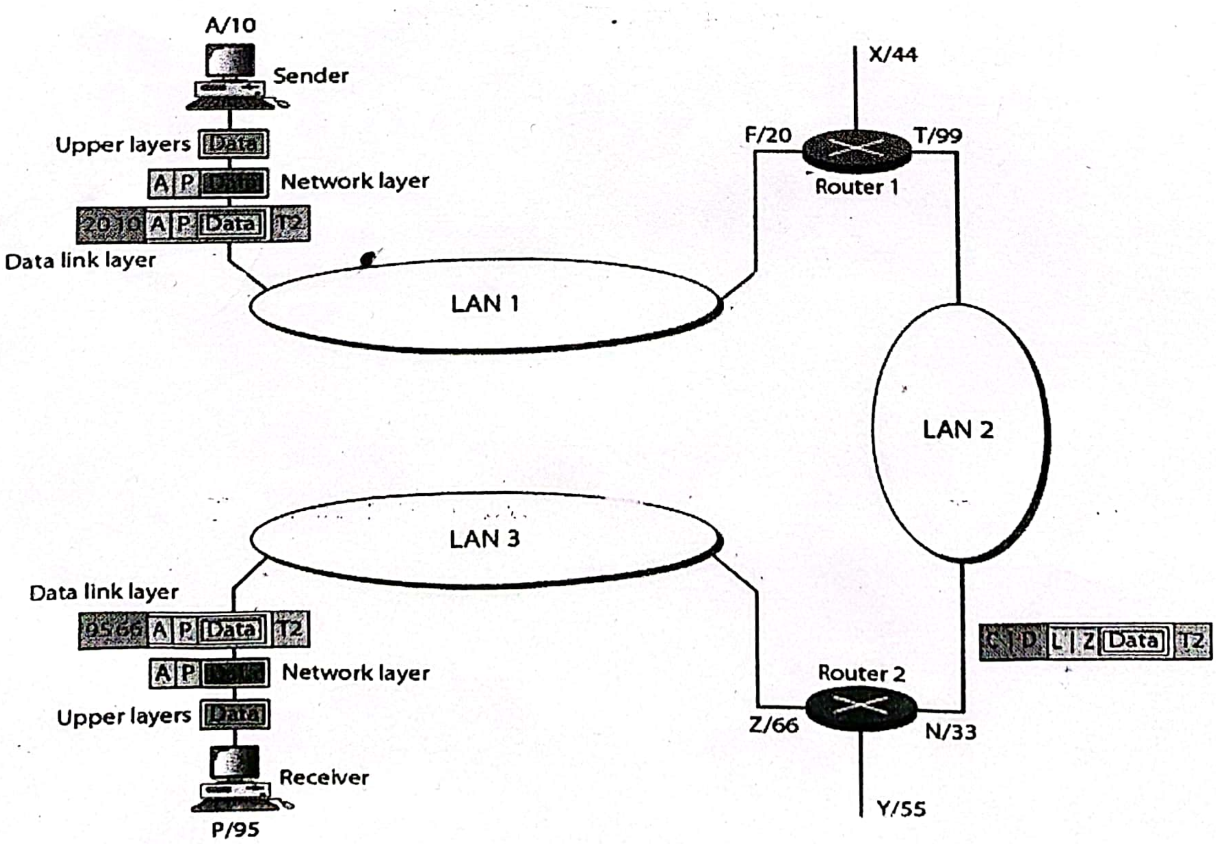

Figure 1: A diagram depicting communication between two devices for Question 1.c

2. a) State the Nyquist bit rate formula. How does the Nyquist bit rate formula differ from the Shannon capacity formula? Consider a channel with 1-MHz bandwidth. The SNR for the channel is 127. What will be the appropriate signal level and bit rate? 10 (CO2) (PO1)

b) Consider the bit stream 0110001001. Draw corresponding digital signals for the following line coding schemes and also comment on the bandwidth requirement of each of the schemes: 3×4 (CO2) (PO1)

i. Pseudoternary

ii. NRZ-I

iii. Manchester

iv. MLT-3

c) Scrambling techniques are used along with line coding schemes to provide synchronization. Scrambling techniques insert pulses to break the long sequences of zeros. Name two widely used scrambling techniques. Consider the bit stream 1100001000000000 and draw the corresponding digital signals by applying the scrambling techniques. How does block coding differ from scrambling? $4 + 2$ (CO2) (PO1)

d) Briefly explain different components of Latency (Delay). A frame of 1 KB size is sent from a source to a destination. The transmission speed of the source is 1 Mbps. A wired medium of CAT5 cable is used for this communication which has an average propagation speed of 2.4×10^8 m/s. There were no intermediate nodes and the distance between the source and the destination is 200 m. Calculate the total delay. $2 + 3$ (CO2) (PO1)

3. a) Differentiate between Multiplexing and Spreading. With necessary diagrams, explain the Frequency Hopping Spread Spectrum (FHSS) technique. $3 + 7$ (CO2) (PO1)

b) Differentiate between synchronous and statistical Time-Division Multiplexing (TDM). Four data channels (digital), each transmitting at 1 Mbps, use a satellite channel (analog) of 1 MHz. Design an appropriate configuration, using Frequency Division Multiplexing (FDM). Briefly explain the modulation technique used in your solution. $4 + 9$ (CO2) (PO1)

c) With necessary diagrams and equations, explain the Pulse Code Modulation (PCM) technique for digitization. 10 (CO2) (PO1)